

Interface Focus

Supplementary Material for

“Red blood cell dynamics in extravascular biological tissues modelled as canonical disordered porous media”

Qi Zhou^{1,*}, Kerstin Schirrmann^{2,3}, Eleanor Doman⁴, Qi Chen^{2,3}, Naval Singh^{2,3}, P. Ravi Selvaganapathy⁵, Miguel O. Bernabeu^{6,7}, Oliver E. Jensen⁴, Anne Juel^{2,3}, Igor L. Chernyavsky^{4,8}, Timm Krüger^{1,*}

¹School of Engineering, Institute for Multiscale Thermofluids, University of Edinburgh, UK

²Manchester Centre for Nonlinear Dynamics, University of Manchester, UK

³Department of Physics and Astronomy, University of Manchester, UK

⁴Department of Mathematics, University of Manchester, UK

⁵Department of Mechanical Engineering, School of Biomedical Engineering, McMaster University, Canada

⁶Centre for Medical Informatics, Usher Institute, University of Edinburgh, UK

⁷The Bayes Centre, University of Edinburgh, UK

⁸Maternal and Fetal Health Research Centre, School of Medical Sciences, University of Manchester, UK

*Corresponding authors: q.zhou@ed.ac.uk (Q.Z.); timm.krueger@ed.ac.uk (T.K.)

S1 Numerical model for cellular flow simulation

The immersed-boundary-lattice-Boltzmann method (IB-LBM [1]) is employed to simulate cellular blood flow as a suspension of deformable RBCs. The numerical model is implemented in the parallel flow simulator *HemeLB* [2] (open source at <https://github.com/hemelb-codes/hemelb>), which has been successfully applied to cellular simulation in complex and sparse geometries (*e.g.*, realistic microvasculature [3]). The 3D fluid flow governed by the Navier-Stokes equations is solved by the lattice-Boltzmann method (LBM) with standard D3Q19 lattice [4], BGK collision operator [5], Guo’s forcing scheme [6] and the Bouzidi-Firdaouss-Lallemand (BFL) implementation of no-slip condition at the walls [7]. Open boundary conditions for flow inlets/outlets are implemented with the Ladd velocity boundary condition [8]. To configure the simulation, each porous medium geometry (generated in STL format) is discretised into a uniform mesh of cubic voxels. The voxel size (*i.e.*, the lattice constant Δx) is chosen such that the plasma flow can be reliably solved at the narrowest pore-throats in the simulated porous media. The relaxation time parameter τ_{BGK} is determined by minimal relative error of simulated flow rates [9].

The discrete RBCs are modelled as triangulated Lagrangian membranes using the finite element method (FEM) and coupled with the fluid flow using the immersed-boundary method (IBM [10]). Each RBC consists of N triangular facets and presents a discocyte shape at rest. The RBC mesh resolution of the membrane is chosen such that the average edge length of its facets matches the voxel size Δx of the flow domain for numerical stability and accuracy [11]. The modelled membrane is hyperelastic, isotropic and homogeneous, with its mechanical properties controlled by elastic moduli (κ_s , κ_b , κ_α) and surface/volume moduli (κ_A , κ_V) as listed in Table S1. κ_{cc} and κ_{cw} are intensities of the repulsive potential (decaying with inverse distance) between neighbouring cells, and neighbouring cell-wall, respectively. The cytoplasm enclosed by the RBC membrane is regarded as a Newtonian fluid with the same viscosity to the plasma. The membrane viscosity is not considered. Table S1 contains values of key simulation parameters for the porous medium simulations presented in the present study.

Table S1. Key parameters for the simulated flow and RBCs. “ $\sim$ ” indicates dimensionless units.

Parameter	Description	Value	Unit	Comment
Δx	lattice constant	0.4	μm	decided by minimal throat widths
Δt	time step length	1.6×10^{-8}	s	based on diffusive scaling
c_s	lattice speed of sound	14.43	m/s	$c_s = \sqrt{\frac{1}{3}} \Delta x / \Delta t$
ρ_0	plasma density	1×10^3	kg/m^3	same as liquid water
η_0	plasma dynamic viscosity	1	mPa s	same as liquid water
ν_0	plasma kinematic viscosity	1×10^{-6}	m^2/s	$\nu_0 = \eta_0 / \rho_0$
τ_{BGK}	relaxation time	0.8	–	satisfying $\nu_0 = c_s^2 (\tau_{\text{BGK}} - \frac{1}{2}) \Delta t$
R_{rbc}	RBC radius	4	μm	radius of biconcave disc
T_{rbc}	RBC thickness	2.5	μm	width of biconcave disc
A_{rbc}	RBC surface area	140	μm^2	conserved during deformation
V_{rbc}	RBC volume	100	μm^3	conserved during deformation
$\eta_{\text{cytoplasm}}$	cytoplasmic viscosity	1	mPa s	simplified as $\eta_{\text{cytoplasm}} = \eta_0$
N	number of membrane facets	2000	–	$N = 20(R_{\text{rbc}}/\Delta x)^2$
Γ	Föppl-von Kármán number	400	–	$\Gamma = \kappa_s R_{\text{rbc}}^2 / \kappa_b$
$\tilde{\kappa}_s$	strain modulus	1.67×10^{-3}	–	physiological reference $5 \mu\text{N/m}$
$\tilde{\kappa}_b$	bending modulus	4.18×10^{-4}	–	physiological reference $2 \times 10^{-19} \text{ J}$
$\tilde{\kappa}_\alpha$	dilation modulus	0.5	–	conserving local area of membrane
$\tilde{\kappa}_A$	surface modulus	1	–	conserving cell surface area
$\tilde{\kappa}_V$	volume modulus	1	–	conserving cell volume
$\tilde{\kappa}_{cc}$	repulsion intensity	$\tilde{\kappa}_b$	–	governing cell-cell interaction
$\tilde{\kappa}_{cw}$	repulsion intensity	$10\tilde{\kappa}_b$	–	governing cell-wall interaction

To validate the described RBC model for simulating cellular blood flow in confined geometries such as the pore-throat network studied in the present work, benchmark tests of a single RBC in narrow cylindrical channels are performed considering a range of channel sizes, flow velocities and cell positions (Fig. S1). Both the temporal evolution and steady state of the simulated cell's deformation agree well with *in vitro* observation of healthy human RBC under comparable flow conditions [12, 13]. The deformability of the simulated RBC is quantified using the capillary number Ca defined as below:

$$Ca = \eta_0 \dot{\gamma}_c R_{\text{RBC}} / \kappa_s, \quad \dot{\gamma}_w = 8\bar{u}/D \quad (\text{S1})$$

where $\dot{\gamma}_w$ is the wall shear rate calculated from the mean flow velocity u and channel diameter D .

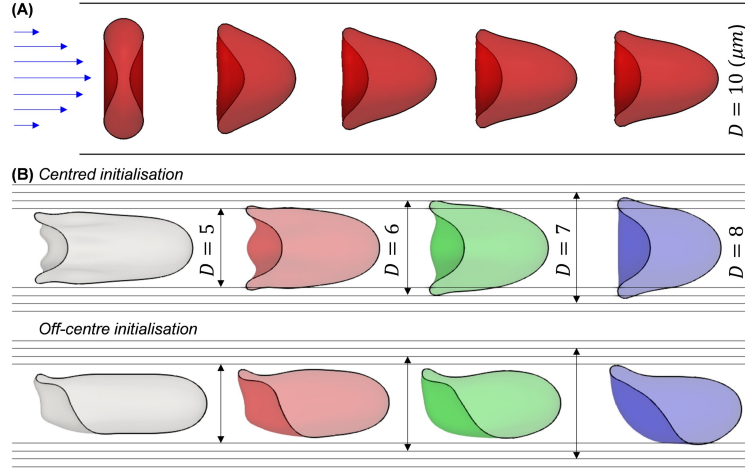


Fig. S1. Benchmark tests of RBC deformation in confined shear flow. (A) Time-lapse deformation of a single RBC in a cylindrical tube of diameter $D = 10 \mu\text{m}$ ($Ca = 2.0$). (B) Steady state of the RBC's deformation in a cylindrical tube of diameter $D = 5, 6, 7, 8 \mu\text{m}$ ($Ca = 0.76, 0.63, 0.54, 0.48$), respectively. The top row corresponds to RBCs initialised at the channel centreline, whereas the bottom row corresponds to RBCs initialised at a slightly off-centre position.

S2 Effect of porosity on flow resistance

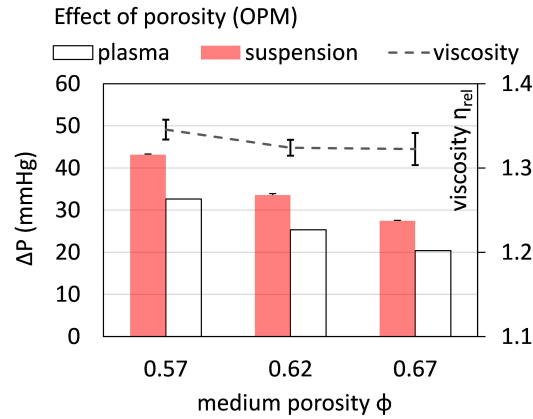


Fig. S2. Supplementary figure for Fig. 3A. Same conventions as Fig. 3A, except for OPMs here.

S3 Empirical model for blood viscosity in tube flow

Capitalising on various sources of experimental data collected from literature, Pries and co-workers [14] derived an empirical model that depicts the variation of the apparent viscosity of blood η_{app} (relative to that of plasma η_0) in cylindrical tubes against the tube diameter D (formulated in μm) and discharge haematocrit H_D :

$$\eta_{\text{rel}} = \frac{\eta_{\text{app}}}{\eta_0} = 1 + (\eta_{45} - 1) \frac{(1 - H_D)^C - 1}{(1 - 0.45)^C - 1} \quad (\text{S2})$$

where η_{45} is the relative apparent viscosity of blood at a discharge haematocrit $H_D = 0.45$ formulated as

$$\eta_{45} = 220e^{-1.3D} + 3.2 - 2.44e^{-0.06D^{0.645}}, \quad (\text{S3})$$

and C is a coefficient related to D in the form of

$$C = (0.8 + e^{-0.075D}) \left(-1 + \frac{1}{1 + 10^{-11}D^{12}} \right) + \frac{1}{1 + 10^{-11}D^{12}}. \quad (\text{S4})$$

S4 Spatiotemporal dynamics of the RBC suspension flow

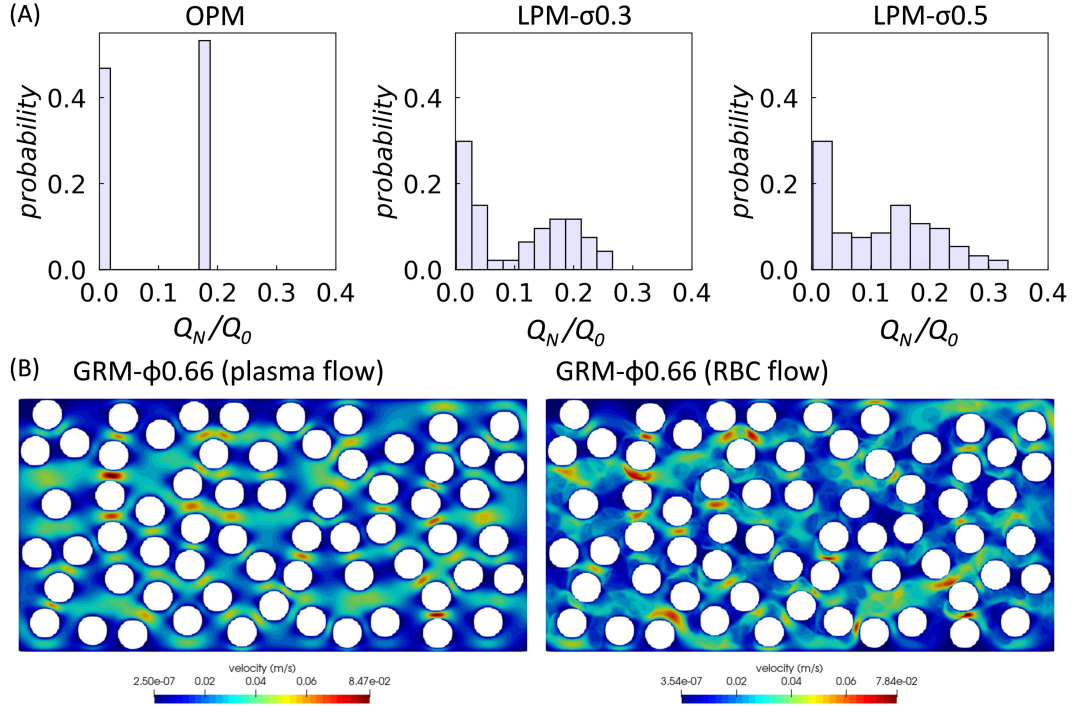


Fig. S3. Effect of RBCs interacting with structural disorder on flow distribution within the porous media. (A) Flow rate distributions for steady plasma flows (Newtonian) through LPM with increasing levels of disorder $\sigma = 0, 0.3, 0.5$ (where $\sigma = 0$ refers to the ordered porous medium), corresponding to the RBC flows Q_S ($H_F = 20\%$) in Fig. 4B. Porosity is $\phi = 0.57$ in all cases. The histograms show the Newtonian flow rates Q_N evaluated at all throats within the pore space network. $Q_0 = 0.4 \mu\text{L}/\text{min}$ is the imposed flow rate at the inlet. (B) Flow channelisation patterns for (left) steady plasma flow and (right) RBC flow ($H_F = 20\%$) in GRM- $\phi 0.66$ under the same inflow $Q_0 = 0.4 \mu\text{L}/\text{min}$.

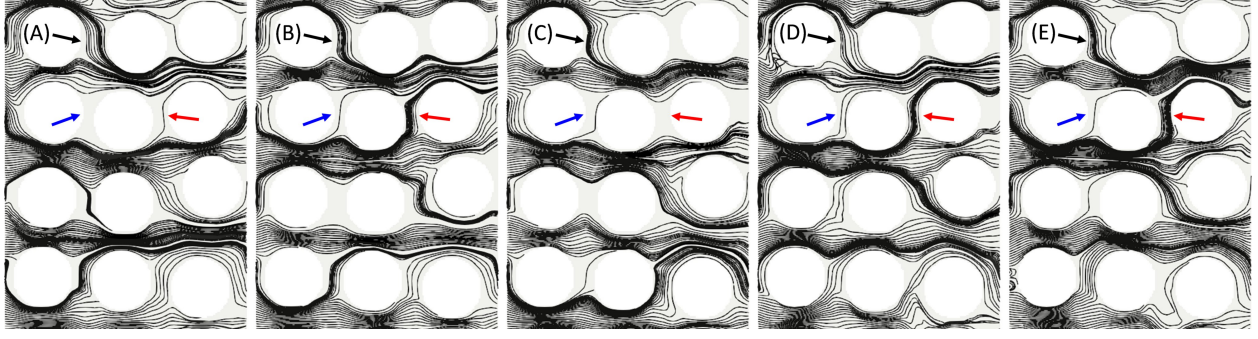


Fig. S4. Instantaneous streamlines of the simulation in Fig. 4A. The corresponding time instants are $t = 134.8$ ms, $t = 137.8$ ms, $t = 140.8$ ms, $t = 143.7$ ms, $t = 146.7$ ms, respectively. The temporal variation of the streamlines is caused by the dynamics of the discrete RBCs. The coloured arrows in (A–E) indicate the three throats of interest (“1”, “2”, “3”) marked in panel Fig. 4A.

S5 Effect of medium porosity and disorder on RBC transport

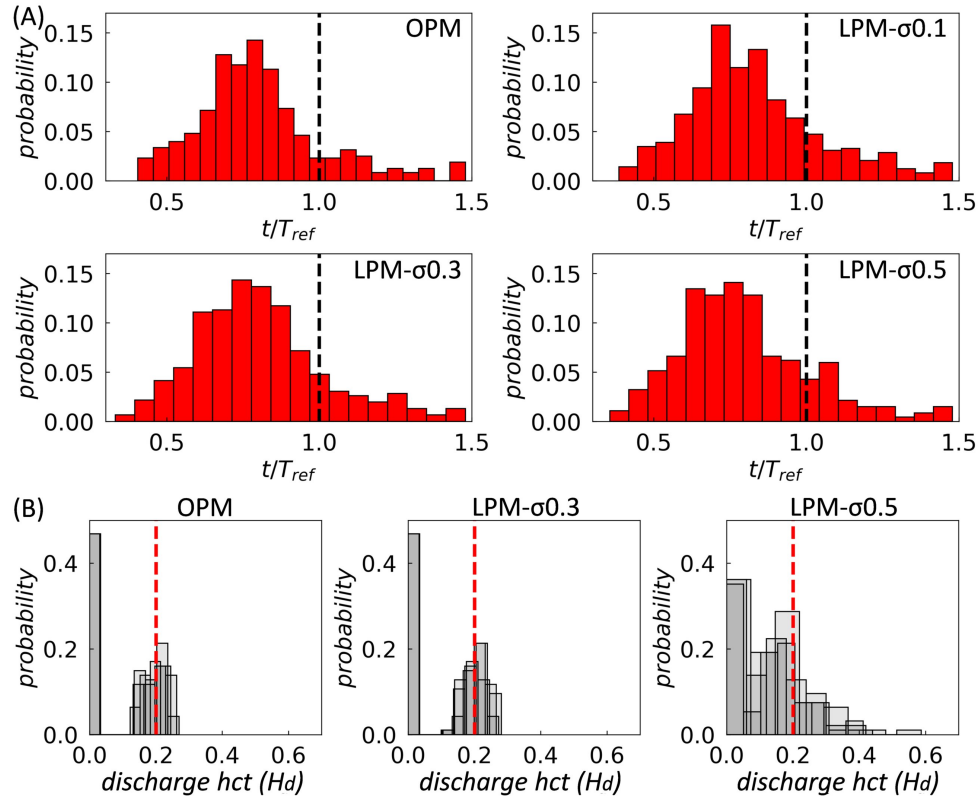


Fig. S5. Quantification of particle transit time and discharge haematocrit in other geometries ($\phi = 0.57$). (A) Transit time distributions in the OPM and LPM geometries ($\sigma = 0.1, 0.3, 0.5$). (B) Discharge haematocrit distributions in the OPM and weakly disordered LPM geometries ($\sigma = 0.1, 0.3$).

S6 Supplementary movies

Movie1 Video clip of the RBC simulation in OPM- $\phi 0.57$ ($H_F = 0.2$).

Movie2 Video clip of the RBC simulation in LPM- $\sigma 0.5$ - $\phi 0.57$ ($H_F = 0.2$).

Movie3 Video clip of the droplet experiment in LPM- $\sigma 0.5$ - $\phi 0.57$ ($H_F = 0.2$).

Movie4 Video clip of the RBC simulation in LPM- $\sigma 0.5$ - $\phi 0.67$ ($H_F = 0.2$).

Movie5 Video clip of the RBC simulation in LPM- $\sigma 0.7$ - $\phi 0.67$ ($H_F = 0.2$).

Movie6 Video clip of the RBC simulation in GRM1- $\phi 0.66$ ($H_F = 0.2$).

S7 Simulation files

The input files for RBC simulations central to the findings of this article, together with cell trajectory data corresponding to the video clips in Sec. S6, are provided in the compressed package SimulationFiles.zip.

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